The prompt engineering strategies shown in figure 8.1 can be applied to memory and knowledge. Knowledge isn't considered memory but rather an augmentation of the prompt from existing documents. Both knowledge and memory use retrieval as the basis for how unstructured information can be queried.

The retrieval mechanism, called retrieval augmented generation (RAG), has become a standard for providing relevant context to a prompt. The exact mechanism that powers RAG also powers memory/knowledge, and it's essential to understand how it works. In the next section, we'll examine what RAG is.

8.2 The basics of retrieval augmented generation (RAG)

RAG has become a popular mechanism for supporting document chat or question-and-answer chat. The system typically works by a user supplying a relevant document, such as a PDF, and then using RAG and a large language model (LLM) to query the document.

Figure 8.2 shows how RAG can allow a document to be queried using an LLM. Before any document can be queried, it must first be loaded, transformed into context chunks, embedded into vectors, and stored in a vector database.